Paderborn Center for Parallel Computing

Accelerating Buffer Sizing for Dataflow Architectures: Control-Path-Driven Black-Box RTL Simulation in FINN

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Automatic FIFO Sizing in DNN accelerator compilers

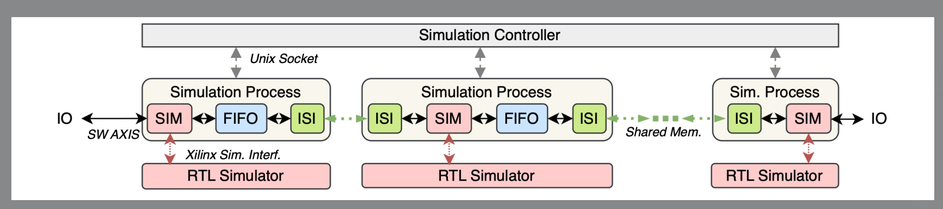
- DNN inference acceleration is an increasingly important FPGA workload
- Dataflow architectures offer a promising alternative to systolic arrays and instruction-based architectures (CPU, GPU, NPU)
- Automatic DNN-to-accelerator compilers are essential for making dataflow architectures practical to use
- Inter-layer buffers (FIFOs) are required for correct and performant dataflow operation

Identifying the Problem

- Sizing FIFOs without simulation requires modeling each layer's timing behavior
  - Modeling is hard: black-box IP can't be modeled, hand-crafted models need expert upkeep
  - Manual modeling is intractable for deep, multi-branch DNN topologies
- Simulation-based approaches are either slow or white-box, HLS-only (no RTL and/or Blackbox support)
- Inference accelerators such as FINN require rapidly iterating a design, slow FIFO sizing is bottleneck
- Requirements for a solution:
  - Correctness (no deadlocks)
  - Performance Guarantee
  - Fast sizing, scalable to large/deep DNNs
  - Minimal Resource Overhead

Simulation Techniques

Distributed Simulation



Overview of the distributed simulation infrastructure. SIM is the simulation interface of the RTL simulation. ISI is the Inter-Simulation Interface.

- Parallel simulation of DNN layers
  - Regular design of DNN allows for efficient distribution
  - One simulation process per layer (incl. MPI support for large DNNs)
  - Replace RTL FIFOs with software FIFOs → faster, more observable

Control-Path Simulation

- Hardware simulation is inherently slow
- NN layer I/O behavior is typically data-independent (data-oblivious scheduling)
- The control path alone determines I/O behavior
  - The data path can be pruned!
- Fixing inputs to constant zero + constant propagation eliminates the data path during synthesis

Hardware-Aware Algorithm

- Fine-granular search of related algorithms inefficient
  - FIFO capacity is allocated in discrete blocks (LUTRAM or BRAM)
- Search at memory-primitive granularity - skips iterations with identical resource / performance outcomes & rejects deadlock/performance degrading configurations
- Simulate with infinite FIFOs to get a safe, performance-optimal upper bound per FIFO (Most FIFOs already sized after this step)
- Minimize each FIFO's depth layer-by-layer via (exponential) binary search over primitive-capacity steps
- Reduces search iterations up to 65× vs. naive binary search

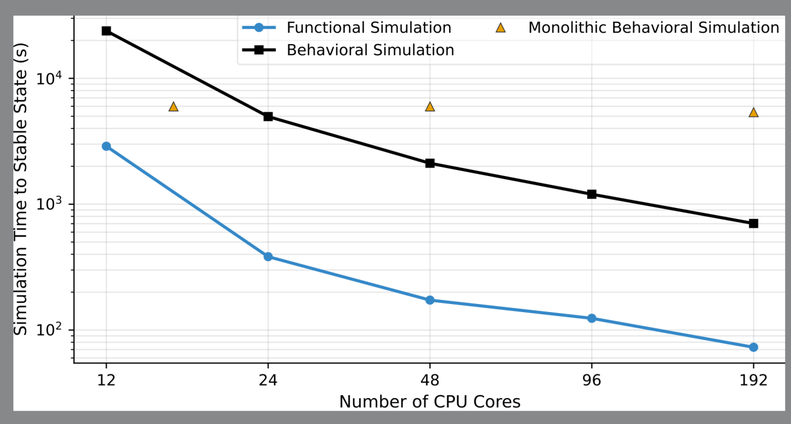
References

[1] F. Jentzsch et al., "LiveFIFO: FPGA-in-the-loop buffer sizing for dataflow accelerators," in Proceedings of the 22nd International Symposium on Applied Reconfigurable Computing (ARC), To appear, 2026

Results

- Same performance guarantee and comparable FIFO sizes compared to best existing approach [1]
- On average 2.88x lower time to solution, no FPGA-in-the-loop required
- No DNN size limit compared to best existing approach

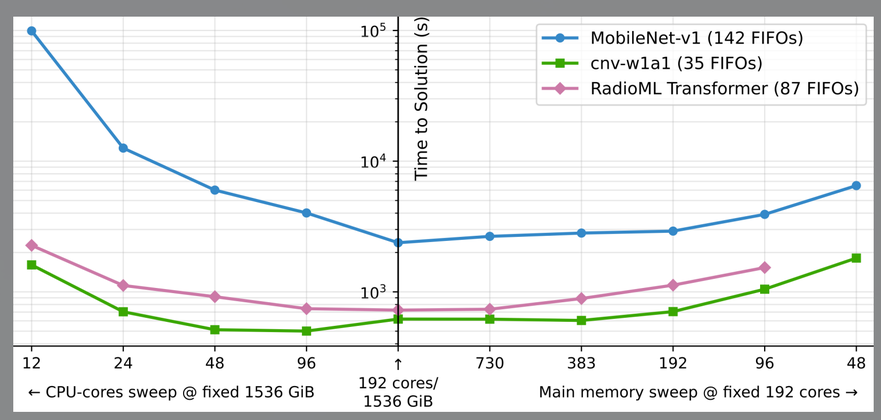
Simulation Improvements



Simulation time to stable state for MobileNetv1 across varying CPU core counts.

- Functional Distributed Control-path simulation vs. monolithic behavioral simulation: ~53.5× faster at 48 cores, ~73.7× faster at 192 cores
- Functional Distributed Control-path simulation provides an order of magnitude of improvement in simulation time vs. Behavioral Distributed simulation

Core / Memory Scaling



Strong scaling behavior of core counts and memory for three selected DNNs.

- Building the simulation scales with available memory (parallel synthesis/compilation)
- Running the simulation scales with CPU cores (avoids per-core process-switching overhead)
- Smaller DNNs are build-time dominated; larger DNNs are simulation-time dominated
- Typical FPGA development workstation is sufficient to size even large models — no cluster needed

Outlook

Our new FIFO Sizing approach is now integrated into the open-source FINN+ compiler and can be used!

FINN+

Check out the repository or try out FINN+ directly:

```
pip install finn-plus
finn build cfg.yaml model.onnx
```

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